

DATA DICTIONARY

FIFA World Cup National Team Performance Analysis (2002–2018)

1. Overview

The Data Dictionary provides a comprehensive description of the datasets used in the FIFA World Cup National Team Performance Analysis project. It documents the tables, fields, data types, relationships, and key analytical measures used throughout the Power BI solution.

The model follows a **Star Schema**, consisting of one fact table, two dimension tables, and one measures table.

2. Data Model Overview

Table Name	Table Type	Purpose
Fact_FIFA	Fact Table	Stores tournament performance statistics for each country.
DimCountry	Dimension Table	Stores unique participating countries.
DimYear	Dimension Table	Stores unique FIFA World Cup tournament years.
Measures	Measures Table	Stores all DAX calculations used in the dashboards.

3. Fact Table Documentation

Table Name: Fact_FIFA

Description:

The Fact_FIFA table contains the quantitative performance metrics for each participating country in each FIFA World Cup tournament from 2002 to 2018. Each row represents a country's performance in a specific tournament.

Column Name	Data Type	Description	Example
Tournament Year	Whole Number	FIFA World Cup tournament year	2014
Position	Whole Number	Final tournament ranking	1
Team	Text	Participating national team	Germany
Games Played	Whole Number	Total matches played	7
Win	Whole Number	Number of matches won	6

Column Name	Data Type	Description	Example
Draw	Whole Number	Number of matches drawn	1
Loss	Whole Number	Number of matches lost	0
Goals For	Whole Number	Total goals scored	18
Goals Against	Whole Number	Total goals conceded	4
Goal Difference	Whole Number	Goals scored minus goals conceded	14
Points	Whole Number	Total tournament points	19

4. Dimension Table Documentation

Table Name: DimCountry

Description:

The DimCountry table contains a unique list of participating national teams and provides descriptive information used for filtering and grouping analyses.

Column Name	Data Type	Description
Team	Text	Name of the participating country

Table Name: DimYear

Description:

The DimYear table contains the distinct FIFA World Cup tournament years included in the project.

Column Name	Data Type	Description
Tournament Year	Whole Number	FIFA World Cup tournament year

5. Relationship Documentation

The data model follows a **Star Schema** with single-directional filtering from dimension tables to the fact table.

From Table	From Column	To Table	To Column	Relationship
DimCountry	Team	Fact_FIFA	Team	One-to-Many
DimYear	Tournament Year	Fact_FIFA	Tournament Year	One-to-Many

6. DAX Measures Documentation

Document only the principal measures that drive your analysis.

Measure Name	Description
Total Countries	Counts the distinct number of participating countries.
Total Games Played	Calculates the total matches played across all tournaments.
Total Wins	Calculates the total number of wins.
Total Draws	Calculates the total number of drawn matches.
Total Losses	Calculates the total number of losses.
Total Goals Scored	Calculates the total goals scored by all teams.
Total Goals Against	Calculates the total goals conceded by all teams.
Total Points	Calculates the cumulative tournament points.
Goal Difference	Calculates the net goal difference.
Goal Change	Calculates the year-over-year change in goals scored.
Highest Tournament Goals	Returns the tournament with the highest total goals scored.
Lowest Tournament Goals	Returns the tournament with the lowest total goals scored.
Tournament Average Wins	Calculates the average wins across participating countries.
Tournament Average Losses	Calculates the average losses across participating countries.
Rank by Goal Difference	Ranks countries according to cumulative goal difference.

7. Data Source Information

Attribute	Description
Source	FIFA World Cup tournament statistics (provided CSV files)
File Format	CSV
Import Method	Power BI Folder Connector
Number of Tournaments	Five (2002–2018)
Data Preparation Tool	Power Query (M Language)
Reporting Tool	Microsoft Power BI Desktop
Data Model	Star Schema

8. Data Quality Summary

The following data quality procedures were applied during the ETL process:

- Combined multiple CSV files into a single fact table.
- Promoted the first row of each file to column headers.
- Replaced the Unicode minus sign in the **Goal Difference** column with the standard negative sign.
- Converted fields to appropriate data types.
- Extracted the tournament year from the source filename.
- Created dimension tables from distinct values.
- Validated relationships and DAX calculations.

9. Business Rules

The following business rules guided the analytical model:

- Each record represents one country's performance in one FIFA World Cup tournament.
- Goal Difference = Goals For – Goals Against.
- Tournament points are taken directly from the source data.
- Rankings are based on cumulative performance measures within the selected filter context.
- All measures respond dynamically to slicers and report filters.

10. Data Model Diagram

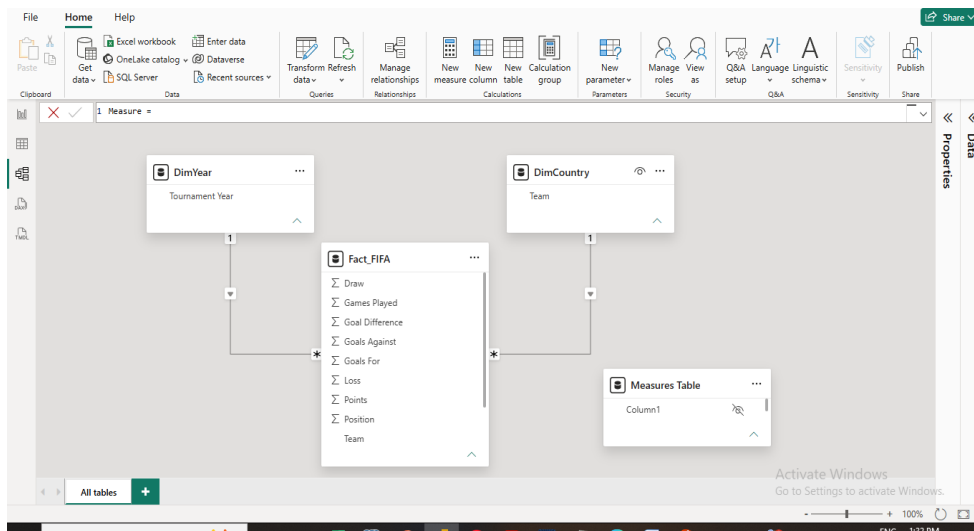


Figure 1. Star Schema Data Model for FIFA World Cup Analysis

The data model consists of one central fact table (Fact_FIFA) connected to two dimension tables (DimCountry and DimYear) through one-to-many relationships. A dedicated Measures table stores DAX calculations separately from the data model, improving readability, maintainability, and report development.